Day 94: Real Generated Text From the Mini-GPT — an Honest Quality Assessment

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 14, Day 4 | Complete study guide

Watch: Nucleus Sampling: The Curious Case of Neural Text Degeneration (TechViz) | Language Model Evaluation and Perplexity (Machine Learning TV)

Script: `day94_generated_text_quality.py` · every number in this guide is copied by program from `day94_run_output.txt`, the real output of that script.

Day 92's mini-GPT writes 120 passages; so does Day 91's Kneser-Ney counter; the real continuations are the reference. Read one at a time, the GPT's text looks like a play. Measured, it is not Shakespeare: a simple character-statistics classifier tells it from the real text 0.725 of the time at temperature 1.0 and 0.825 with top-p 0.9 (two halves of the real text: 0.442, chance). It does not copy its training text any more than real unseen text shares phrases with it by chance, and it writes too few speaker lines (0.144 of lines against 0.260).

0. Where today fits

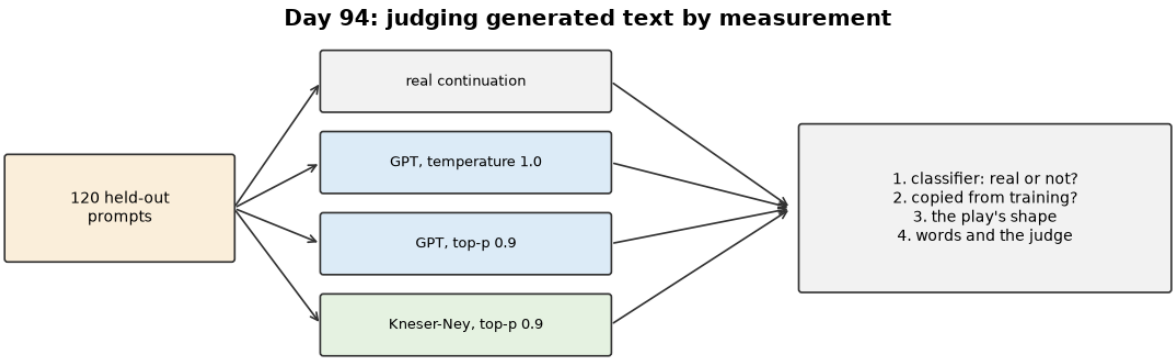
DEFINITION — start here, no prior knowledge assumed

Judging generated text by reading a few samples is unreliable: people remember the good ones. A **quality assessment** replaces impressions with measurements on many samples, each against a **reference** computed the same way on real text, and with a **control** that shows what 'no difference' looks like.

REAL-WORLD SCENARIO

A wine tasting done blind, with the same wine poured twice as a check on the tasters.

SYSTEM DIAGRAM



Four kinds of passage from the same 120 prompts, four tests.

Today's part for Day 97

`day94_exercise.py` is the set of text measures (copying, distinct pairs, real words, speaker lines), each checked on texts whose answer is known in advance.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
Why this test	The question the experiment answers.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. Four kinds of passage

DEFINITION — start here, no prior knowledge assumed

120 prompts of 32 characters from random places in the held-out text. After each: what Shakespeare really wrote next, the GPT at temperature 1.0 and at top-p 0.9, and Kneser-Ney (6 characters of context) at top-p 0.9; 300 characters each.

REAL-WORLD SCENARIO

Four writers finishing the same 120 opening lines.

ACTUAL OUTPUT

```
the first 200 characters after prompt 1 ('ely gait?\nO, be thou Dian, and l'):
--- real ---
et her be Kate;
And then let Kate be chaste and Dian sportful!
KATHARINA:
Where did you study all this goodly speech?
PETRUCHIO:
It is extempore, from my mother-wit.
KATHARINA:
A witty mother! witl
--- GPT, temperature 1.0 ---
et he shall go.
KING RICHARD II:
O, pardon this: good moyates as I have a palace?
Why, fair now, I may now it, let he day at like,
Thou art York anoral, as i' the next from him,
As I had doing thee p
--- GPT, top-p 0.9 ---
et he shall her be it sends
no desperate harmours here thy way,
When at name fair truth, and some about now
Is think had that hath conveit a band,
Such do you send some doth consenting hate.
More spea
--- Kneser-Ney, top-p 0.9 ---
ived of your valiant, lies
With mine is little behold
Dish;
And child were I am going them off my hands;
Which we will prove me this finger, do your way service.
Provost:
A partial set my fair, he di
Actual output from this lesson's run.
```

What it means

Read alone, the GPT's passages have the look of a play: speaker names, verse lines, Shakespearean words. Some words are invented ('moyates', 'conveit'), and the sense drifts within a line. Kneser-Ney looks similar at a glance. Reading cannot say how similar; the next sections measure it.

WHY CHOOSE THIS?

Shows what the numbers below are about.

WHY NOT THIS? (TRADEOFFS)

One example per source; the measurements use all 120.

2. Can a simple classifier tell it from real Shakespeare?

DEFINITION — start here, no prior knowledge assumed

Each passage becomes counts of its 1-, 2- and 3-character pieces (hashed into 4,096 columns). A logistic regression, written from scratch, learns real against generated, and is scored on passages it did not train on (five-fold cross-validation). 0.5 means it cannot tell.

REAL-WORLD SCENARIO

A blind test with a careful but simple taster, and the same wine poured twice as the check.

MATHEMATICS

$$p(\text{generated} \mid x) = 1 / (1 + e^{-(w \cdot x + b)}); \quad \text{minimise } \text{mean}[-y \log p - (1 - y) \log(1 - p)] + (\lambda/2)|w|^2$$

IMPLEMENTATION

```
def logistic_cv(Xa, Xb, seed=0):
    X = np.vstack([Xa, Xb]); y = np.r_[np.zeros(len(Xa)), np.ones(len(Xb))]
    folds = np.random.RandomState(seed).permutation(len(y)) % FOLDS
    acc = []
    for f in range(FOLDS):
        tr, te = folds != f, folds == f
        w, b = np.zeros(X.shape[1]), 0.0
        for _ in range(CLF_STEPS):
            p = 1 / (1 + np.exp(-(X[tr] @ w + b)))
            g = p - y[tr]
            w -= CLF_LR * (X[tr].T @ g / tr.sum() + L2 * w); b -= CLF_LR * g.mean()
        acc.append(np.mean(((X[te] @ w + b) > 0) == y[te]))
    return np.mean(acc), np.std(acc)
```

ACTUAL OUTPUT

```
real against                accuracy  spread over folds
real (other half)              0.442              0.073
GPT, temperature 1.0           0.725              0.053
GPT, top-p 0.9                 0.825              0.101
Kneser-Ney, top-p 0.9          0.754              0.050
Actual output from this lesson's run.
```

What it means

The control works: two halves of the real text give 0.442, chance. The GPT's text is told apart 0.725 of the time at temperature 1.0 and 0.825 with top-p 0.9. A likely reason, not tested here: cutting off the unlikely tail makes the text more typical of the model and less like real text, which is full of unlikely characters. Kneser-Ney's is told apart 0.754 of the time. The classifier only sees short character pieces; one that read whole sentences would very likely do better, so these accuracies are a lower bound on how detectable the text is.

WHY CHOOSE THIS?

A number, with a control, instead of an impression.

WHY NOT THIS? (TRADEOFFS)

A weak detector: it gives a lower bound on detectability, not a quality score.

3. How much is copied?

DEFINITION — start here, no prior knowledge assumed

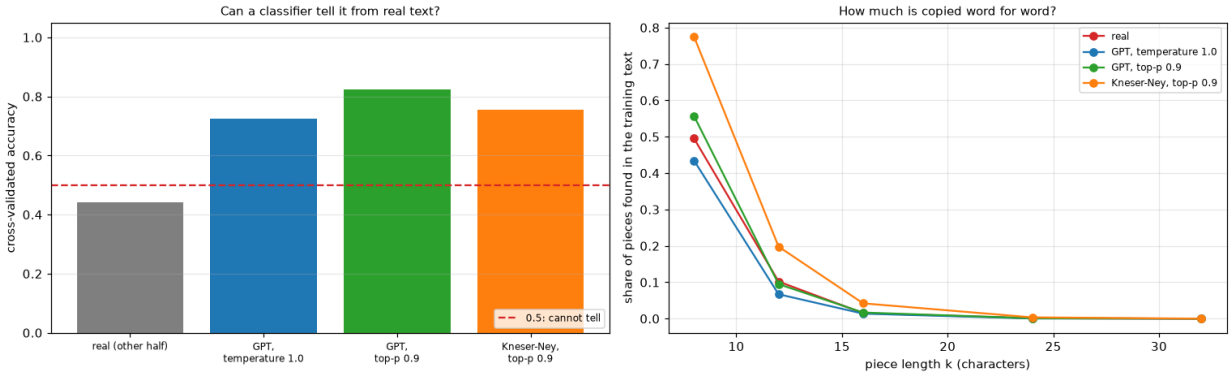
For each piece length k, the share of all k-character pieces of the passages found word for word in the training text. Real held-out text is the reference: Shakespeare repeats phrases, so some sharing is expected by chance.

REAL-WORLD SCENARIO

Checking an essay against a library: some common phrases always match.

ACTUAL OUTPUT

```
source                          k=8     k=12     k=16     k=24     k=32
real                          0.495    0.102    0.015    0.001    0.000
GPT, temperature 1.0          0.434    0.067    0.014    0.000    0.000
GPT, top-p 0.9                0.557    0.095    0.017    0.001    0.000
Kneser-Ney, top-p 0.9         0.775    0.198    0.042    0.004    0.000
Actual output from this lesson's run.
```



Left: classifier accuracy (0.5 = cannot tell). Right: the share of k-character pieces found in the training text.

What it means

The GPT copies about as much as real unseen text shares by chance: at 16 characters 0.014 and 0.017 against 0.015, and nothing of 32. It writes new text, not quotations. Kneser-Ney copies more (0.775 of 8-character pieces, 0.042 of 16-character ones), likely because with only 6 characters of context it often has a single seen continuation, so it follows the training text.

WHY CHOOSE THIS?

A direct test for memorisation, against the right reference.

WHY NOT THIS? (TRADEOFFS)

Exact matches only; close paraphrases are not caught.

4. The play's shape, and the words

DEFINITION — start here, no prior knowledge assumed

Speaker lines (a name in capitals followed by a colon, alone on a line), average line length, and Day 93's measures: distinct word pairs, real words, and the Kneser-Ney judge's bits per character.

REAL-WORLD SCENARIO

Checking a forged script for the right layout as well as the right vocabulary.

ACTUAL OUTPUT

```
source                    speaker lines  line length  distinct pairs  real words  judge bits
real                             0.260         30.3           0.758       0.926       2.214
GPT, temperature 1.0             0.144         34.9           0.890       0.876       2.382
GPT, top-p 0.9                   0.142         36.6           0.819       0.941       2.040
Kneser-Ney, top-p 0.9            0.240         33.6           0.805       0.969       1.453
Actual output from this lesson's run.
```

What it means

The GPT writes too few speaker lines (0.144 and 0.142 of lines against 0.260) and longer lines (34.9 and 36.6 characters against 30.3): its speeches run on. Its real-word rate is close to the real text's (0.876 and 0.941 against 0.926; the real text likely has rare names the training text lacks). The judge's bits favour Kneser-Ney's own text (1.453), as expected, since the judge is the same counting model; between the GPT and the real text it is a fair comparison (2.382 and 2.040 against 2.214).

WHY CHOOSE THIS?

Measures that point at specific weaknesses (layout, run-on speeches).

WHY NOT THIS? (TRADEOFFS)

Simple measures; none captures meaning.

Interview preparation — technical questions

Q: How do you check a text detector is not fooling itself?

A: A control: real against real should give chance (0.442 here).

Q: How do you test whether a model memorised its training data?

A: Share of long pieces found verbatim in training, against real unseen text as the reference.

Q: Why hash the classifier's features?

A: A fixed number of columns for any number of distinct pieces.

Q: Why cross-validate the classifier?

A: Accuracy on passages it trained on would overstate how separable the sources are.

Q: Why is the judge unfair to compare Kneser-Ney with itself?

A: It is the same model; its own text is what it expects.

Interview preparation — theoretical questions

Q: Why was top-p text easier to detect than temperature 1.0?

A: Measured here (0.825 against 0.725); a likely reason is that cutting the unlikely tail makes text more typical than real text.

Q: Does a low detection accuracy mean good text?

A: No: it only means this detector failed; stronger detectors may succeed.

Q: Why measure copying against real held-out text?

A: Real text also shares common phrases with the training text; that is the baseline.

Q: What does the shape test reveal that bits per character do not?

A: Structural errors: too few speaker changes, speeches that run on.

Key takeaway and what's next

> **KEY TAKEAWAY**
> Measured against real text with controls, the mini-GPT's writing is new (it copies no more than chance) and plausible at a glance, but detectable by a simple character classifier 72-83% of the time, and its plays have too few speaker changes. Reading samples alone would have hidden all of this.

- Classifier: control 0.442; GPT 0.725 (temperature 1.0), 0.825 (top-p 0.9); Kneser-Ney 0.754.
- Copying of 16-character pieces: real 0.015, GPT 0.014-0.017, Kneser-Ney 0.042.
- Speaker lines: real 0.260, GPT 0.144 and 0.142.
- Line length: real 30.3, GPT 34.9 and 36.6 characters.

Suggested watch order, if starting from zero

- 1. Language Model Evaluation and Perplexity (Machine Learning TV) — what perplexity measures, and what it does not.
- 2. Nucleus Sampling: The Curious Case of Neural Text Degeneration (TechViz) — why generated text degenerates and how sampling changes it.

Day 95 preview

- **Scaling.** How the held-out loss moves with model size and with the amount of training text, and how far four points can be trusted.